MAKING A RESOLUTION ASSIGNMENT INTO A FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7653-7674, following the statement at 7637-7651.

Editorial note. No omissions: the uniqueness delegation is to fully carded lemmas; functoriality follows from uniqueness.

Allowed standard prerequisites (editorial).

- Lifting and uniqueness against injective resolutions. lemma-morphisms-lift (carded L26169) and lemma-morphisms-equal-up-to-homotopy (carded L26190), or equivalently lemma-morphisms-into-injective-complex (carded L26222), all cited at ~7652. Existence and uniqueness of j(alpha) making the square commute in K^+.

Source reference. Original lemma 013S: sources/stacks-derived/pinned-original-source.tex, source lines 6698–6698. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TG: sources/stacks-derived/pinned-original-source.tex, source lines 6758–6758. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013P: sources/stacks-derived/pinned-original-source.tex, source lines 6603–6603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough injectives. Given a resolution functor (j, i) there is a unique way to turn j into a functor and i into a 2-isomorphism producing a 2-commutative diagram*

$$\begin{array}{ccc} K^+(\mathcal{A}) & \xrightarrow[j]{} & K^+(\mathcal{I}) \\ & \searrow \quad \swarrow & \\ & D^+(\mathcal{A}) & \end{array}$$

where $\mathcal{I}$ is the full additive subcategory of $\mathcal{A}$ consisting of injective objects.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. For every morphism $\alpha : K^\bullet \to L^\bullet$ of $K^+(\mathcal{A})$ there is a unique morphism $j(\alpha) : j(K^\bullet) \to j(L^\bullet)$ in $K^+(\mathcal{I})$ such that

$$\begin{array}{ccc} K^\bullet & \xrightarrow{\alpha} & L^\bullet \\ {\scriptstyle i_{K^\bullet}}\downarrow & & \downarrow{\scriptstyle i_{L^\bullet}} \\ j(K^\bullet) & \xrightarrow{j(\alpha)} & j(L^\bullet) \end{array}$$

is commutative in $K^+(\mathcal{A})$. To see this either use Lemmas 013P and 013S or the equivalent Lemma 05TG. The uniqueness implies that j is a functor, and the commutativity of the diagram implies that i gives a 2-morphism which witnesses the 2-commutativity of the diagram of categories in the statement of the lemma. $\square$